

Progressive gate-unfreezing of the QFT operator

One gate thawed per stage and trained to a stationarity plateau — three unfreeze orderings $\times$ two initialisations $\times$ three datasets

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Question

The `QFTBasis(m, m)` used here has every gate trainable on its $U(2)$ manifold but its operator starts either at the QFT-family **identity** ($H \rightarrow I_2$, controlled-phase $\rightarrow$ phase 0, so $T_{t=0}(x) = x$) or at a **Haar-random** point. Rather than training all gates at once, we **unfreeze one gate per stage** and train each stage to a plateau before thawing the next. Two questions: (i) does the **order** in which gates are thawed matter — block-growth (**bg**), left $\rightarrow$ right (**lr**), right $\rightarrow$ left (**rl**)? and (ii) does the **initialisation** — identity vs random — change the trajectory or the endpoint?

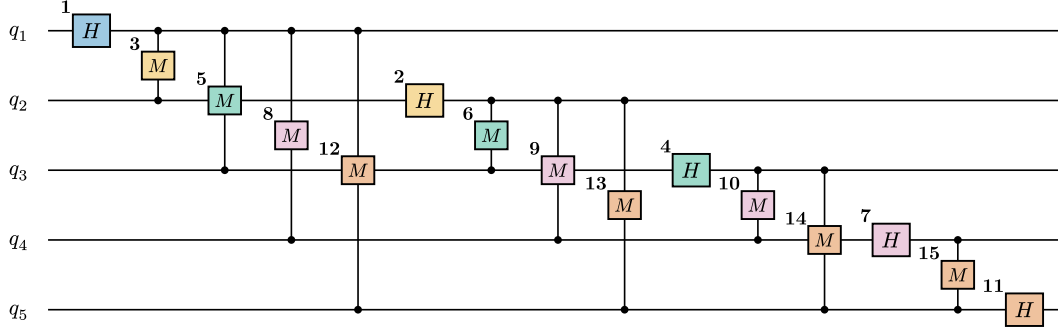
Setup

Each run optimises a **fixed batch** of the training split (full $n = 500$ at QuickDraw $m = 5$; batch 50 at $m = 8$) under an **MSELoss** that keeps the top-10% of coefficients by magnitude. The loop drives pdf’s JIT-compiled Adam step with a per-stage frozen set — Adam moments reset between stages, carried within a stage, constant learning rate — and a separate Riemannian grad-norm probe supplies the plateau signal. A stage ends when $\|g\| < 10^{-5}$ **or** $|\Delta L| < 10^{-5}$ (after a 5-step minimum), or at the step cap (800 at $m = 8$). `QFTBasis(m, m)` has $m(m+1)$ gates, so the sweep runs 30 stages at $m = 5$ and 72 at $m = 8$. The three orderings are fixed index sequences into the Hadamard-first gate storage; random init draws a Haar $U(2)$ on each H slot and a uniform phase on each controlled-phase slot, with a seed shared across the three orderings so they start from the same basis.

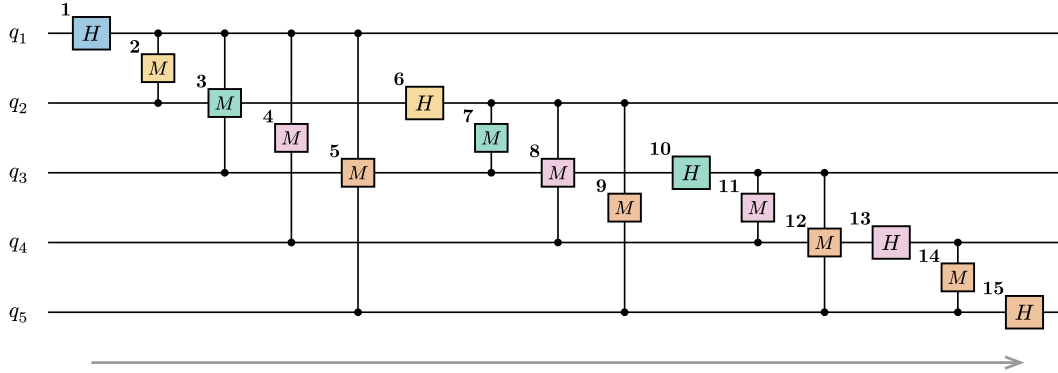
QFT gate-unfreezing — three orderings (single axis, $m = 5$)

Same QFT(5) circuit in the paper's tensor-network notation (controlled-phase M = box between two wires with legs to a dot on each). Bold badge = **unfreeze step** for that ordering; fill = block-stage k (= highest qubit a gate touches). Cumulative: each step thaws one more gate, the rest stay frozen at identity. (No SWAP gates — bit-reversal is folded into the index convention.)

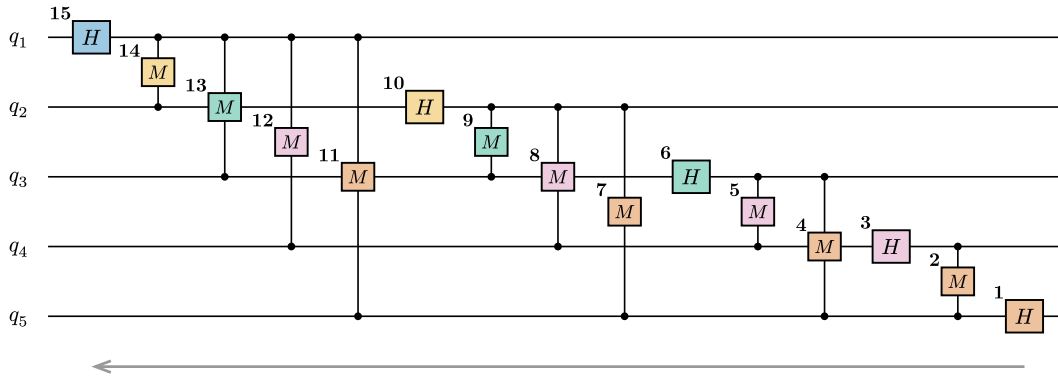
(a) block-growth stage k completes QFT(k, k); long-range couplings delayed — the gate-level mirror of the block sweep



(b) left → right sweep unfreeze in QFT construction order 1→15; q_1 couples to all of q_2 – q_5 before H_2 at step 6



(c) right → left sweep reverse construction order; last-built gate (H_5) first, H_1 last



Block-stage colours: $k = 1$ $k = 2$ $k = 3$ $k = 4$ $k = 5$

Figure 1: The three unfreeze orderings on the QFT(5) circuit (paper tensor-network notation; controlled-phase M = box spanning two wires). The **bold badge** on each gate is its unfreeze step for that ordering; the fill colour is the block-stage k (the highest within-axis qubit the gate touches). **(a) block-growth** — stage k completes QFT(k, k); long-range couplings are delayed (the gate-level mirror of the block-size sweep). **(b) left→right** — QFT construction order 1–15, so q_1 couples to all of q_2 – q_5 before H_2 at step 6. **(c) right→left** — reverse construction order; last-built gate (H_5) first, H_1 last.

Training dynamics

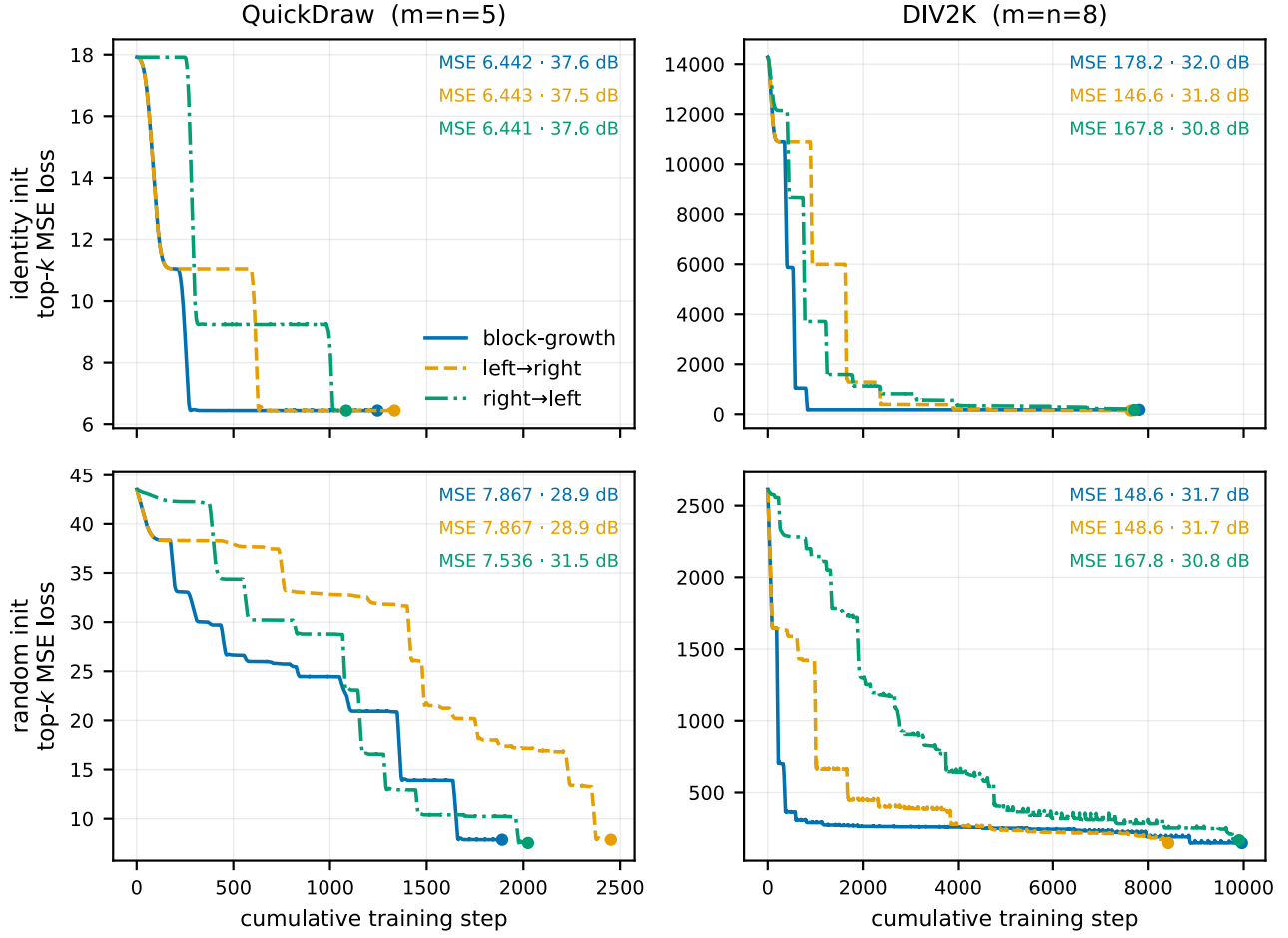


Figure 2: **Absolute** top- k MSE loss (not a normalised ratio) versus cumulative optimiser step, one curve per unfreeze ordering. Rows: identity (top) vs Haar-random (bottom) init; columns: datasets. Each curve is a staircase — every downward step-and-plateau is one freshly-thawed gate trained to stationarity; the final MSE and PSNR@ $\rho\{=\}.20$ are marked at each curve’s end. The per-(dataset, init) two-panel staircases (MSE **and** Riemannian grad norm) are in `<dataset>/<init>/figures/staircase.{pdf,svg}`.

Final reconstruction PSNR

Test-split PSNR (dB) once the last gate has been thawed and trained, at four keep ratios ρ , with the cumulative step count and the per-stage plateau triggers (grad-norm / loss- Δ / step-cap):

QuickDraw (m=n=5, 30 gates)

init	order	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.15$	$\rho\{=\}.20$	steps (g/ Δ /cap)
identity	block-growth	17	22.8	29.1	37.6	1246 (0/30/0)
identity	left→right	17	22.9	29.1	37.5	1334 (0/30/0)
identity	right→left	17	22.8	29.1	37.6	1084 (0/30/0)
random	block-growth	17.4	21.6	25.3	28.9	1890 (0/30/0)
random	left→right	17.4	21.6	25.3	28.9	2452 (0/30/0)
random	right→left	17.3	22	26.5	31.5	2024 (0/30/0)

DIV2K (m=n=8, 72 gates)

init	order	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.15$	$\rho\{=\}.20$	steps (g/ Δ /cap)
identity	block-growth	19.4	26.9	29.7	32	7810 (0/72/0)
identity	left→right	25.2	27.9	30	31.8	7635 (0/72/0)

init	order	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.15$	$\rho\{=\}.20$	steps (g/ Δ /cap)
identity	right→left	24.8	27.2	29.1	30.8	7709 (0/72/0)
random	block-growth	25.2	27.8	29.8	31.7	9962 (0/72/0)
random	left→right	25.2	27.8	29.8	31.7	8415 (0/72/0)
random	right→left	24.8	27.2	29.1	30.8	9904 (0/72/0)

Reference: best PSNR from prior experiments

The strongest test-PSNR achieved by **any** prior trained basis on each dataset (the `family_init_matrix` sweep: five circuit families $\times$ two inits $\times$ block sizes $k = 2..8$), as a ceiling to read the gate-unfreeze numbers against:

dataset	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.20$
QuickDraw	18.9 dB	24.1 dB	49.6 dB
DIV2K	26.4 dB	29.3 dB	33.7 dB
TU-Berlin	37.3 dB	53.4 dB	88.2 dB

Best method per cell at $\rho\{=\}.20$: QuickDraw 49.6 dB (`qft/identity`, $k=1$) DIV2K 33.7 dB (`rich/identity`, $k=3$) TU-Berlin 88.2 dB (`mera/random`, $k=2$). Note these maxima sit at **small block sizes** on the two sparse drawing sets (QuickDraw, TU-Berlin) — the near-identity small-block basis is already an excellent sparse representation there, which is exactly the regime where the gate-unfreeze PSNR peaks **early** (see Reading).

Comparison: block-size progressive baseline

The `qft_progressive` experiment reaches the **same** full QFT(8, 8) operator on DIV2K by a different curriculum — a **block-size** schedule that trains the entire $2^k \times 2^k$ QFT (all $k(k+1)$ gates at once, 1008 steps) at each stage $k = 1..8$, instead of thawing one gate at a time. It is the natural reference for the question **does the curriculum/order matter for the final operator, or only for the path?**

stage k	block	# gates	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.15$	$\rho\{=\}.20$
1	2×2	2	9.4	12.5	16.2	21.1
2	4×4	6	19.1	26.9	29.8	32
3	8×8	12	25.2	28.1	30.3	32.3
4	16×16	20	25.3	28	30.1	32
5	32×32	30	25.2	27.8	29.8	31.7
6	64×64	42	25.2	27.8	29.8	31.7
7	128×128	56	25.2	27.8	29.8	31.7
8	256×256	72	25.2	27.8	29.8	31.7

DIV2K test PSNR (dB) per keep ratio ρ

The block-size curriculum **saturates by $k = 2$** and the full QFT(8, 8) endpoint is **31.7 dB** at $\rho\{=\}.20$ (25.2 / 27.8 / 29.8 dB at $\rho = .05, .10, .15$; = the all-gates-at-once `qft_identity` result). Read this against the **DIV2K** rows of the gate-unfreeze table above: if the identity-init sweep also lands at ≈ 31.7 dB, the QFT(8, 8) optimum is **schedule-independent** — neither the unfreeze order nor the one-gate-at-a-time vs whole-block curriculum changes the destination, only the trajectory (exactly what the QuickDraw panels show, where all three orderings converge). A gate-unfreeze endpoint **below** the block-size baseline would instead say the per-gate schedule gets trapped short of the joint optimum.

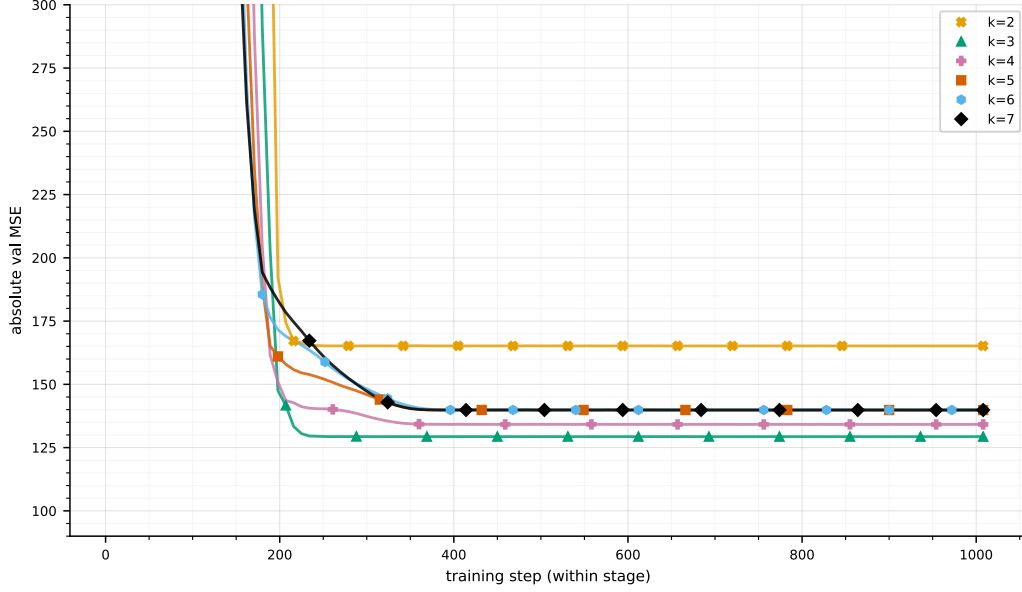


Figure 3: Reference (`qft_progressive`, DIV2K-8q): the block-size curriculum’s per-stage validation MSE, $k = 2..7$. Eight whole-block stages here vs the 30/72 per-gate stages of the unfreeze staircases above — same endpoint, very different path granularity.

Reading

The staircases (absolute top- k MSE) make the per-gate marginal contribution legible: a tall step is a gate that mattered, a flat plateau a gate the optimiser left near where it was thawed. Compare the three orderings within each panel for **path** effects and the two init rows for **initialisation** effects. On QuickDraw the training MSE converges to the same floor for all three orderings within an init (order changes only the path), while identity reaches a **lower** MSE than random — the QFT-family identity carries a usable inductive bias even under a one-gate-at-a-time schedule. The trigger mix (grad-norm vs loss- Δ vs step-cap) reports **how** each stage terminated.

Training loss $\neq$ test PSNR on sparse data. The optimiser minimises the top- k MSE in the **transform** domain, which is not the same objective as test-image PSNR. On QuickDraw the per-stage test PSNR@ $\rho\{=\}.20$ is **highest early** — up to **54.8 dB** around **stage 5**, near the identity/pixel operator — and then **falls to ≈ 37.6 dB** as more gates train the operator toward the QFT and spread energy away from the sparse-pixel representation. So for sparse drawings the PSNR-optimal point is **early in the unfreeze schedule**, not at the fully-trained end; this matches the reference table above, where the prior-experiment maxima also sit at small block sizes. (Per-stage PSNR is recorded in each cell’s `trace.json`; on DIV2K, where images are not pixel-sparse, training instead **improves** PSNR toward the ≈ 31.7 dB block-size baseline.)